High Validity, Low Transfer: Anchored Evaluation of Diffusion Counterfactual Explanations for Chest Radiographs

Supplementary Material

R. Reproducibility Appendix

Every value in this appendix is read from the code and configuration of the default-configuration runs of the main paper (2-model guidance ensemble) and checked against their run logs, recorded configuration and checkpoints. Settings that differ between code versions are listed in Sec. R5 with the value the default runs used.

R1. Reward

Notation: x is the episode's original image (fixed for the episode), x_{t+1} the image after policy step t, $d_t = |f(x_t) - c|$ with f the guidance classifier, S the soft lung mask of x (computed once per episode, Table R4), and $p_O(\cdot)$ the softmax, clamped to $[10^{-8}, 1]$, of the frozen YOLOv8n feature vector (Table R4). Means and sums run over the 256×256 pixels. The per-step reward is

$$r_t = k_c \sum_j \lambda_j R_j,$$

with $k_c = 2.25$ for n→p ($c = 1$) and $k_c = 1$ for p→n ($c = 0$), applied to every term.

$\lambda_{\text{sp}}(u) = 0$ outside the sparsity arm. In that arm it is 0 for updates $u < \text{round}(0.2U)$, rises linearly to 2.0 at $u = \text{round}(0.6U)$ and stays there ($U = 100$ updates). A safeguard measures CV every 10 updates, from 10 updates before the ramp until its end, on 24 held-out far-pool images with the deterministic policy and a budget of $\min(4, T)$ steps. If CV falls more than 0.08 below its pre-ramp value, it freezes λ_{sp} for the rest of training at the value of the last probe that passed. In the trained sparsity arm the safeguard froze λ_{sp} at 0 for n→p at update 20, before the ramp moved; p→n ramped to 2.0.

R2. Action mapping

The actor outputs the mean μ of a 9-dimensional diagonal Gaussian over a raw, unbounded vector $a \in \mathbb{R}^9$. During training a is sampled; at evaluation $a = \mu$. Each control is a fixed transform of a, with σ the logistic sigmoid (Table R2). The raw vector is never clipped. Ranges are open intervals, because σ never reaches 0 or 1.

Discrete controls. No Bernoulli distribution is used. a_4 and a_5 are Gaussian like the other dimensions and are thresholded deterministically, $e_i = \mathbf{1}[\sigma(a) > 0.5] = \mathbf{1}[a > 0]$, identically in training and evaluation. The PPO log-probability is that of the raw Gaussian sample, with no correction for the sigmoid or the threshold. Stochasticity in e_1, e_2 comes only from Gaussian exploration during training.

Per-pass schedule. Each policy step runs $n = 2 + e_1 + e_2$ chained generator passes, each taking the previous pass's output as its input. Pass $k = 0, \ldots, n-1$ uses

$$s_k = \min\big(0.9,\ s\,\beta\,(1 + 0.15k)\big),$$
$$g_{c,k} = g_c\, b\,(1 + 0.35k),$$
$$g_{p,k} = g_p\ \max(1/b,\ 0.6)\ \max(1 - 0.15k,\ 0.4),$$

and the same ramp start r. The applied strength s_k therefore ranges over $(0.17, 0.9]$ rather than Table 1's $(0.20, 0.70)$. Neither floor in $g_{p,k}$ binds: $1/b > 0.769$ and $1 - 0.15k \geq 0.55$ for $k \leq 3$.

R3. PPO

Action statistics. From an evaluation replay with the policy's mean action, not read from the saved counterfactuals. In n→p, 199 of 200 episodes end after one step; strength sits at its lower bound and the pass count is three on every step, guidance bias is within 1% of its upper bound on 97.5% of steps, and strength boost is near-constant, while the classifier and preservation scales, stop signal and ramp start vary between images. In p→n the median episode is one step and the mean 5.46, 31 episodes reach the 25-step cap, and over the 71 multi-step episodes 67 to 78% of the action variance is between images.

R4. Generator, LoRA, and background lock

R5. Settings that differ between code versions

Table R5 lists every setting above that differs between code versions, with the value used by the default runs. The 3-model, sparsity and Grad-CAM runs came from uncommitted code states, identified by their configuration hash and logged flags.

Also recorded for the default runs: the patient-level split (80% of patients for training and 20% for test, seed 0, persisted with a dataset fingerprint; one image per patient, 21,347 training and 5,337 test images, unstratified); the random seeds (Table R3); the LHR implementation and the per-shard configuration stamp that fixes the transfer-ratio denominator (Sec. R6); and a run manifest for every training and evaluation run, giving its code revision and configuration. The random-action and no-policy arms compared with the single-classifier policy are evaluation-only and reuse that run's generator, `rsna` guidance and evaluation groups. The no-policy, fixed-schedule and second-seed runs compared with the default (main-paper Sec. 5.2,

Table R1. Reward terms as implemented. B is an L_1 edit budget on validity; with $B = \infty$ validity depends on d_{t+1} alone.

Term	Implemented formula	Weight λ_j
R_{prog} (progress)	$d_t - d_{t+1}$	1.5
R_{val} (distance)	$-d_{t+1}$	2.0
R_{close2} (squared distance)	$-d_{t+1}^2$	2.5
R_{fid} (fidelity)	$-\operatorname{mean}_p \lvert x_{t+1} - x\rvert$ (mean pixel L_1)	2.0
R_{pres} (preservation)	$-\Big(\frac{\sum_p S\,\lvert x_{t+1} - x\rvert}{\sum_p S + 10^{-8}} + \eta\,\mathrm{KL}\big(p_O(x)\,\|\,p_O(x_{t+1})\big)\Big)$, $\eta = 0.5$; KL summed over feature dimensions	3.0
R_{bud} (per-step cost)	-1	0.1
R_{hit} (validity bonus)	$\mathbf{1}\big[d_{t+1} < \epsilon_{\text{stop}} \wedge \operatorname{mean}_p \lvert x_{t+1} - x\rvert < B\big]$, $\epsilon_{\text{stop}} = 0.08$, $B = \infty$; validity also ends the episode, so this is paid at most once, on the last step	2.0
R_{stop} (stop bonus)	$R_{\text{hit}} \cdot \mathbf{1}[\tau_t > 0.5]$, $\tau_t = \sigma(a_3)$	2.0
R_{sp} (sparsity arm only)	$-\operatorname{mean}_p \lvert x_{t+1} - x\rvert$; $(1 - \text{CAM})$-weighted and normalized by the weight sum when a Grad-CAM mask is active	$\lambda_{\text{sp}}(u)$

Table R2. Raw-action-to-control mapping, checked cell by cell against Table 1 of the main paper.

Dim.	Control	Mapping	Range	Main-paper Table 1
a_0	DDIM strength s (base value)	$0.20 + 0.50\,\sigma(a_0)$	$(0.20, 0.70)$	matches
a_1	classifier-guidance scale g_c	$0.30 + 0.90\,\sigma(a_1)$	$(0.30, 1.20)$	matches
a_2	preservation-guidance scale g_p	$0.30 + 1.20\,\sigma(a_2)$	$(0.30, 1.50)$	matches
a_3	stop signal τ	$\sigma(a_3)$	$(0, 1)$	matches
a_4, a_5	extra passes e_1, e_2	$\mathbf{1}[\sigma(a) > 0.5]$	$\{0, 1\}$	matches
a_6	strength boost β	$0.85 + 0.30\,\sigma(a_6)$	$(0.85, 1.15)$	matches
a_7	guidance bias b	$0.70 + 0.60\,\sigma(a_7)$	$(0.70, 1.30)$	matches
a_8	ramp start fraction r	$0.10 + 0.40\,\sigma(a_8)$	$(0.10, 0.50)$	matches

5.4; Supp. E5) instead use the default's guidance, evaluation groups, images and per-image seeds, with each direction's own adapter (the second seed's two directions share one adapter). The held-out transfer point estimates were recomputed from the saved evaluation shards and agree with the originally logged values to within 10^{-4}; the differences come from GPU type and affect only the last printed digit. The bootstrap intervals reported in the main paper come from this recomputation (Sec. R6).

R6. Metric definitions

Notation: $X_n = \{x : f(x) < 0.2\}$ and $X_p = \{x : f(x) > 0.8\}$ are the first $N \le 200$ images of each kind, in dataset order, selected by the guidance classifier f rather than by label. f is the ensemble's mean probability in ensemble arms. n→p edits X_n and p→n edits X_p, with $s = +1$ and $s = -1$ respectively. Images are grayscale, 256×256, in $[0, 1]$, and $\delta = \lvert x_c - x\rvert$ per pixel.

Counterfactual Validity.

$$\mathrm{CV} = \frac{1}{N}\sum_i \mathbf{1}\big[s\,(f(x_c^i) - f(x^i)) \ge \tau \wedge B_i\big], \quad \tau = 0.8,$$

with B_i: $f(x^i) \le 0.5 < f(x_c^i)$ for n→p and $f(x^i) \ge 0.5 > f(x_c^i)$ for p→n. Pairs with a non-finite score are dropped. Mean CV is the unweighted mean of the two directions. On X_n and X_p the margin condition already implies B_i: with $f \in [0, 1]$, $s\,(f(x_c^i) - f(x^i)) \ge 0.8$ forces $f(x_c) \ge 0.8$ for n→p and $f(x_c) \le 0.2$ for p→n, and the source side holds by construction of the groups. The tightening relative to a direction-agnostic CV therefore comes from the sign requirement alone.

Fraction of pixels changed. $\phi_i = \frac{1}{HW}\sum_p \mathbf{1}[\delta_p > 0.01]$, averaged over pairs. Pairs with $\sum_p \delta_p = 0$ are skipped, both here and for edit concentration.

Edit concentration. $C_i = \sum_{p \in \mathrm{top}_k(\delta)} \delta_p \big/ \sum_p \delta_p$ with $k = \mathrm{round}(0.05 \cdot 65{,}536) = 3{,}277$, averaged over pairs. Spatially uniform change scores exactly 0.05. As an illustrative heuristic only, not a threshold, a compact lesion-shaped edit would be expected to put roughly half or more of its mass in the top 5% of pixels. We do not use 0.5 as a decision boundary.

Table R3. PPO settings. One policy per direction, trained independently with identical settings.

Setting	Value
Episodes per direction	800
Episodes per update	8 (100 updates)
Minibatch	full batch: one gradient step per epoch on all transitions of the update ($\leq 8T$); no minibatching
PPO epochs per update	4
Optimizer	AdamW, learning rate 3×10^{-4}, constant; other arguments at PyTorch defaults
Clip ϵ	0.2
GAE	$\gamma = 0.99$, $\lambda_{\text{GAE}} = 0.95$; no bootstrap past episode end; advantages normalized to zero mean and unit variance over the update
Value loss	weight 0.5; MSE to the shaped target $r_t + 0.2\,A_t$ if $\tau_t \leq 0.5$, else r_t, where A_t is the unnormalized GAE advantage (not the GAE return)
Entropy bonus	weight 0.01; entropy of the raw 9-d Gaussian, summed over dimensions
Auxiliary heads	weight 0.1 each. Feasibility: sigmoid, BCE to $\mathbf{1}[\hat{R}_t > 0]$ ($\hat{R}_t$ the GAE return). Progress: linear, MSE to $r_t + \gamma V_{t+1}$ ($V_{t+1} = 0$ at episode end). Stop: sigmoid, BCE to $\mathbf{1}[\tau_t > 0.5]$
Saturation penalty	$0.01 \cdot \text{mean ReLU}(\lvert\mu\rvert - 3)^2$ over raw actor means
Gradient clipping	global norm 0.5
Policy std	state-independent learnable per-dimension $\log\sigma$, initialized to -0.5; σ floored at 10^{-3}
Trunk	MLP 73,731 $\to$ 512 $\to$ 256, ReLU; 37,885,462 policy parameters in total. The alternative convolutional trunk has 197,910
Heads	linear from the 256-d trunk feature: actor (9), value (1), feasibility, progress, stop (1 each)
Observation	$[z_x,\ z_{x_t},\ t/T,\ d_t,\ c,\ S(x_t)]$: two VAE-posterior samples ($2 \times 4{,}096$), 3 scalars, and the 256^2 lung mask of x_t; 73,731 entries
Step budget T	25, in training and evaluation
Termination	$d_{t+1} < \epsilon_{\text{stop}} = 0.08$ (with $B = \infty$) or $t = T$
Training images	drawn uniformly with replacement, one per episode, from training-fold images with $\lvert f(x) - c\rvert > 0.5$, excluding the first 24 of that pool, which are held out for the sparsity-arm probe in every arm
Final policy	weights after the last update; no checkpoint selection
Evaluation	mean action ($a = \mu$), same T and termination; per-image seed = base seed 0 + global image index ($+10^6$ for p$\to$n)
Random seeds	split 0; LoRA fine-tuning 0; PPO 1 (n$\to$p) and 2 (p$\to$n), i.e. base 0 plus a per-direction offset; cuDNN deterministic mode on, full determinism off, so repeated runs are not guaranteed to be bit-identical, and re-running the identical protocol on different hardware regenerates different images, so aggregate metrics move by small amounts (two runs of the default p$\to$n protocol differ in whole-image LHR lift by $+0.016$, 95% CI $[-0.045, +0.098]$, on 148 images); evaluation base seed 0

Identity preservation. Both scores are computed per pair (source image vs. its counterfactual) and averaged. *MS-SSIM*: five scales with 2×2 average pooling between them; an 11×11 Gaussian window, $\sigma = 1.5$, zero padding; $C_1 = 0.01^2$ and $C_2 = 0.03^2$ (data range 1); scale weights $(0.0448, 0.2856, 0.3001, 0.2363, 0.1333)$. The result is $\prod_{j=1}^{4} \text{cs}_j^{w_j} \cdot \text{ssim}_5^{w_5}$, with each factor clamped at $\geq 10^{-6}$. *HF-corr*: $h(x) = x - G_{11,1.5} * x$, r the Pearson correlation of $h(x)$ and $h(x_c)$ over all pixels, reported as $(1 + r)/2$.

FID. $\text{FID}(A, B) = \|\mu_A - \mu_B\|^2 + \text{Tr}\big(\Sigma_A + \Sigma_B - 2(\Sigma_A\Sigma_B)^{1/2}\big)$. Here Σ is the unbiased sample covariance plus $10^{-6}I$, and the real part of the matrix square root is taken. Features are the global-average-pooled output (1024-d) of the `features` module of TorchXRayVision DenseNet-121 `densenet121-res224-rsna`, applied to the image rescaled to $[-1024, 1024]$ and resized bilinearly to 224^2. The main paper reports the cross-group values $\text{FID}(X_p, X_c^{\text{n}\to\text{p}})$ and $\text{FID}(X_n, X_c^{\text{p}\to\text{n}})$, where the "real target class" is the classifier-selected group, not the label. The feature extractor is the `rsna` model, which also guides generation in every trained arm (the 3-model ensemble is `rsna`, `chex`, `mimic_nb`), so FID is not independent of guidance.

Held-out transfer ratio. For validator $v \in \{\texttt{nih}, \texttt{pc}, \texttt{mimic_ch}\}$ (TorchXRayVision DenseNet-

121, Pneumonia output) and direction s,

$$\rho_v = \frac{\frac{1}{N}\sum_i s\left(f_v(x_c^i) - f_v(x^i)\right)}{\frac{1}{N}\sum_i s\left(f_g(x_c^i) - f_g(x^i)\right)},$$

over the same N images. f_g is the guidance classifier (mean probability over members for an ensemble), and ρ_v is undefined when the denominator is exactly zero. No validator is a guidance member in any reported arm. The denominator must use the classifier(s) that actually guided generation: it is computed from the guidance roster in each evaluation shard's configuration stamp, which records the guidance flag and member list for the default and later arms; the single-classifier arm's shards predate this key and are treated as single-classifier guidance, which its training log confirms. The 3-validator mean in main-paper Tables 2 and 4 is the unweighted mean of the per-validator ratios. No threshold is applied to ρ_v; it is read against the real-pair ceiling and the attack floor of Supp. E8.

Bootstrap intervals for ρ_v. 10,000 resamples, with replacement, of the $N = 200$ image indices of a group. Numerator and denominator are recomputed on the same resampled indices, and the 95% interval is taken from the 2.5th and 97.5th percentiles.

Lesion Hit Rate. Defined for p→n only, on images with an RSNA opacity annotation. A_i is the union of the image's boxes, rescaled from the native DICOM size to the 256^2 grid and rounded outward to whole pixels. L_i is the source image's lung mask: TorchXRayVision PSPNet lung probability (maximum over the two lung channels, computed as for the lung mask S of Table R4) thresholded at 0.5. With $\delta = |x_c - x|$, the reported in-lung lift is $\ell_i = h_i/\pi_i$ with $h_i = \sum_{A_i \cap L_i} \delta / \sum_{L_i} \delta$ and $\pi_i = |A_i \cap L_i|/|L_i|$, and LHR is the mean of the per-image ratios, not a ratio of pooled sums; images with $\pi_i = 0$ (box entirely outside the lung mask) or with $|A_i|/(HW) \in \{0, 1\}$ are skipped (none of the 148 default-arm images). The whole-image lift replaces L_i by the full image; the editable-region and soft-lung variants of Supp. E7 replace the indicator of L_i by the soft weights m or S in all four sums. The reported t and p for lift against chance are from a one-sample t-test of ℓ_i against 1. Paired arm comparisons use the images both arms share: paired t-test and Wilcoxon signed-rank test on per-image lift differences, d = mean/standard deviation (ddof 1) of those differences, and a 10,000-resample percentile bootstrap CI of the mean difference.

Table R4. Generator settings. x_a is the anchor image: the episode's original image.

Setting	Value
Backbone	SD v1.5 (`runwayml/stable-diffusion-v1-5`) VAE and UNet, fp16, frozen except LoRA. Text encoder unused. 256^2 grayscale input replicated to 3 channels and mapped to $[-1, 1]$; $4 \times 32 \times 32$ latent drawn from the VAE posterior (sample, not mean)
Conditioning MLP	$[c,$ latent average-pooled to 64$] \in \mathbb{R}^{65} \to 256 \to 768$, SiLU; one cross-attention token; 214,272 parameters, fp32, trained jointly with LoRA
LoRA	rank 8, $\alpha = 16$, dropout 0.05, on `to_q`, `to_k`, `to_v`, `to_out.0` of the attention layers (1,594,368 LoRA parameters)
LoRA fine-tuning	ϵ-prediction MSE only; timestep uniform over the 50 DDIM timesteps; 30 epochs, batch 8, AdamW with learning rate 10^{-4}; class-balanced sampling with replacement; fp16 autocast with gradient scaling; gradient-norm clip 1.0. Training images: the 21,347 images of the training fold. Each direction fine-tunes its own adapter with this recipe and seed, and the resulting weights differ. As the Frobenius norm of the difference over all LoRA tensors, relative to the n→p adapter's norm: 0.385 (default), 1.18×10^{-2} (single-classifier), 3.94×10^{-3} (3-model), 1.49×10^{-3} (sparsity), and 0 for the Grad-CAM arm, whose two adapters are identical; the default arm's difference is 33 to 259 times the other arms'. The default and Grad-CAM arms are evaluated with each direction's own adapter, the other arms with the n→p adapter for both directions. All default-arm numbers in the main paper come from the evaluation with each direction's own adapter. An evaluation of the default with the n→p adapter for both directions, on the same images and seeds, is used only against the three arms evaluated that way (main-paper Table 4); against it, the per-direction evaluation has in-lung LHR lower by 0.028 on 148 images ($[-0.058, 0.000]$) and p→n CV lower by 0.045 (19 vs. 10 discordant images, exact McNemar $p = 0.14$); its n→p CV is 0.945 vs. 0.960 (3 vs. 0 discordant, $p = 0.25$). Two runs of the identical matched-adapter protocol, partly on different GPU types, differ by $+0.016$ in whole-image LHR lift ($[-0.045, +0.098]$) and by 0.015 in p→n CV (6 vs. 9 discordant, $p = 0.61$), so the changes are not attributed to the adapter alone
DDIM	SD1.5 scheduler configuration, $T_{\text{diff}} = 50$ timesteps; pass k keeps the last $\max(1, \lfloor 50\, s_k \rfloor)$ and starts from the pass input's latent, forward-noised to the first kept timestep. DDIM $\eta = 0$ (not set in code; library default)
Guidance losses	$\mathcal{L}_{\text{cls}} = \text{BCE}(f(D(\hat{z}_0)), c)$; $\mathcal{L}_{\text{pres}} = \dfrac{\sum S\,\lvert D(\hat{z}_0) - x_a\rvert}{\sum S + 10^{-8}} + \eta\,\text{MSE}\big(O(D(\hat{z}_0)), O(x_a)\big)$, $\eta = 0.5$ (the reward uses KL on softmaxed features instead, Table R1)
Guidance update	$z_{t-1} \leftarrow z_{t-1} - w(i)\, m \odot \big(1.15\, \alpha_{\text{cls}}\, g_{c,k} \nabla_{\hat{z}_0} \mathcal{L}_{\text{cls}} + \alpha_{\text{pres}}\, g_{p,k} \nabla_{\hat{z}_0} \mathcal{L}_{\text{pres}}\big)$, $\alpha_{\text{cls}} = 4.0$, $\alpha_{\text{pres}} = 0.05$; gradients through the VAE decoder D, one backward pass per loss
Trade-off in Eq. 1	classification vs. preservation, no single fixed weight: set per pass by $1.15\, \alpha_{\text{cls}}\, g_{c,k}$ against $\alpha_{\text{pres}}\, g_{p,k}$ (above), and in the reward by λ_{fid} and λ^R_{pres} (Table R1)
Guidance ramp	$w(i) = \text{clip}\big((i - \lfloor nr \rfloor)/\max(1, n - \lfloor nr \rfloor),\ 0,\ 1\big)$ over the n kept steps $i = 0, \ldots, n-1$; no gradient is taken where $w(i) = 0$; its maximum, $(n - 1 - \lfloor nr \rfloor)/(n - \lfloor nr \rfloor)$, is below 1
Lung mask S	TorchXRayVision PSPNet (`chestx_det`) on the image resized to 512^2 and rescaled to $[-1024, 1024]$; sigmoid, maximum over the left- and right-lung channels, resized bilinearly to 256^2
Object features O	YOLOv8n (`yolov8n.pt`) layers 0–9, global average pooled, frozen, on the 256^2 image replicated to 3 channels
Edit mask m	$S(x_a)$ resized bilinearly to 32^2; dilated by a 5×5 stride-1 max-pool (2 latent cells, ≈ 16 px); feathered by a separable Gaussian, $\sigma = 1$ latent cell, radius 3, zero padding; clamped to $[0, 1]$, not thresholded
Background lock	after every step, $z \leftarrow m \odot z + (1 - m) \odot \tilde{z}_a$, where $\tilde{z}_a$ is the latent of x_a forward-noised to the next timestep with fresh noise, and used clean after the last step
Grad-CAM arm only	$m \leftarrow m \cdot \big(0.15 + 0.85\, \widetilde{\text{CAM}}^{\,0.75}\big)$; CAM at layer `features`, normalized by its 0.99 quantile at 32^2, Gaussian blur $\sigma = 0.8$. The trained Grad-CAM arm uses `features.denseblock3`

Table R5. Settings that differ between code versions. "Default runs" is the value in the tables above.

Setting	Default runs	Other code versions
Guidance classifier f	`rsna+chex` mean	`rsna` only (single-classifier arm)
Step budget T	25	8
Episode end	validity or $t = T$	$t = T$ only
L_1 budget B in validity	∞	0.02
Policy trunk	MLP	convolutional
LoRA rank / targets	8 / attention	16 / + `ff.net.0.proj`, `ff.net.2`, `proj_in`, `proj_out`
LoRA epochs	30	15
a_1 mapping	$0.30 + 0.90\,\sigma(a_1)$	$0.50 + 2.50\,\sigma(a_1)$
R_{fid}, λ_{fid}	mean L_1, 2.0	MSE, 0.5
λ^R_{pres}	3.0	0.5
λ_{hit}, λ_{stop}	2.0, 2.0	6.0, 8.0
Lock / preservation anchor	episode's original image	previous pass's output
Saturation penalty	0.01	absent
24-image probe holdout	removed from training pool	absent

E. Additional Details

E1. Controls the comparison requires

A control-arm comparison is only as trustworthy as the plumbing beneath it, and three preconditions we had assumed did not hold during development. (i) A single dataset object was shared by LoRA fine-tuning, the PPO pool and the metrics scan, so evaluation images had been seen during fine-tuning. We replaced it with one deterministic, persisted, *patient-level* split shared by every stage, computed once because generation runs as independent array processes that cannot recompute an RNG-based split consistently. It assigns 80% of patients to training and 20% to test (seed 0; one image per patient in RSNA, so 21,347 training and 5,337 test images) and is stored with a fingerprint of the dataset. In every reported arm, generator fine-tuning, PPO rollouts and the validity probe use only the training fold, and every evaluated image is in the test fold. (ii) Training was unseeded: two identical submissions produced two different policies, which is survivable for a headline number but not for a comparison in which arms must differ by treatment alone. Every reported run now uses split seed 0, LoRA seed 0 and PPO seeds 1 (n→p) and 2 (p→n); evaluation uses base seed 0, with per-image seed equal to the base plus the image's global index ($+10^6$ for p→n). Seeding fixes the random streams but not floating-point nondeterminism, so repeated runs are not guaranteed to be bit-identical. (iii) The control arms had never been exercised. Wiring them up surfaced a routing bug under which every reduce silently fell back to the trained arm's shards, which would have made "the policy contributes nothing" an artefact of a missing flag, and a shared output directory under which each arm overwrote the previous one's metrics.

A CPU-only invariant suite pins the LHR calibration, split disjointness and three further properties whose silent failure would yield a plausible-looking wrong number rather than a crash. Every training and evaluation run writes a manifest giving its code revision and configuration, and every evaluation shard carries a configuration stamp from which the transfer-ratio denominator is read (main paper Sec. 4.3). We report these because each was, at the time it was found, invisible in the outputs: all three would have produced a well-formed table.

E2. Anatomical proxy metrics

We report two anatomical proxies under the default configuration: the Cardio-Thoracic Ratio (CTR), the ratio of cardiac to thoracic width from the heart and lung segmentation masks, and a costophrenic-recess score (SCP) approximating pleural blunting, each tested with a paired t-test on the within-image delta (real minus counterfactual).

CTR shifts by $+0.0127$ ($t = 3.882$, $p = 1.4 \times 10^{-4}$) in n→p and by $+0.0047$ ($t = 0.478$, $p = 0.634$) in p→n: a small shift, significant in n→p only, with an absolute magnitude (≤ 0.013) far below the 0.5 at which a CTR is conventionally read as enlarged. This is consistent with the intended behaviour of the background lock, since the cardiac silhouette lies largely outside the lung mask that gates editing; we read it as a shift too small to interpret either way, not as a directional CTR effect. SCP's own computation flags itself as degenerate under this configuration (relative spread 0.19% across the four image groups), so its delta and t-test are withheld rather than reported on what is, numerically, a near-constant signal. Since it is built on the feature norm of a COCO-pretrained detector rather than a dedicated recess detector, we do not consider it informative in its current form. Neither proxy supports a directional claim about pleural or cardiac status.

E3. Baselines under the same protocol

Published values are not comparable to ours: they are computed with other classifiers, feature extractors and evaluation groups, and none of ECED, FiRe or DocVCE reports a lesion-anchored metric or the held-out target shift. DiME, ACE and FastDiME guide a pixel-space diffusion model trained on the target dataset, FiRe a Pixel Mean Flow model, LDCE a class-conditional latent diffusion model, and DVCE-style methods an adversarially robust classifier. None of these exists for RSNA at our resolution and split, and training one per method was not feasible before submission. We therefore ran the two baselines we could run under an identical protocol: the same 400 test images, groups, guidance ensemble, CV criterion, validators and scoring code as the default policy (Table E1).

Latent Shift (Cohen *et al.*, MIDL 2021; main-paper reference list) walks the latent of a pretrained autoencoder (TorchXRayVision ResNetAE `101-elastic`) along the guidance gradient. The original single-step version, a line search along one gradient direction up to $4\|z_0\|$, plateaus at a guidance shift of 0.3–0.65 and reaches CV 0.005/0.12. We therefore also ran an iterative version that re-computes the gradient at every step (step $0.02\|z_0\|$, at most 150 steps) and stops at the first step that satisfies main-paper Eq. (4), mirroring the policy's stop-at-validity rule. Its hyperparameters were fixed before any result was seen and nothing is tuned on LHR or transfer. **CycleGAN** is an unconditional, unguided image-to-image translation trained between the radiologist-labelled classes of the training fold, applied with one deterministic forward pass per image.

Three observations. First, iterative Latent Shift reproduces the high-CV, low-transfer pattern with an unrelated generator: CV 1.00 in n→p with mean transfer 0.153, and `nih` 0.039, the same as the policy's 0.044. So the gap is not specific to diffusion or to the RL scheduler, though it may still be specific to gradient-guided editing with our guidance models. Second, its in-lung LHR is lower than the policy's,

Table E1. Baselines on the default policy's 400 test images. Transfer: mean ρ_v over three validators (ensemble denominator). LHR: in-lung lift [95% CI], whole-image lift in parentheses; paired in-lung difference to the default on the 148 annotated images.

	Default policy	Latent Shift (iter.)	Latent Shift	CycleGAN
CV n→p / p→n	0.945 / 0.670	1.000 / 0.995	0.005 / 0.120	0.000 / 0.000
Transfer n→p / p→n	0.281 / 0.276	0.153 / 0.224	—	—
`nih` n→p / p→n	0.044 / 0.165	0.039 / 0.187	—	—
Guidance shift n→p / p→n	0.898 / 0.761	0.833 / 0.819	0.567 / 0.677	0.123 / 0.055
LHR (in lung)	1.165 [1.12, 1.21]	1.106 [1.08, 1.14]	1.055 [1.01, 1.10]	1.160 [1.12, 1.20]
(whole image)	(2.595)	(1.451)	(1.174)	(1.483)
default − baseline	—	$+0.059$, p=0.004	$+0.110$, p=1.5×10^{-4}	$+0.005$, p=0.84
Edit mass in lung (p→n)	0.66	0.37	0.30	0.36

but only by 0.059; under the whole-image null the gap reads 1.144, because Latent Shift puts only 37% of its edit mass in the lung. Third, CycleGAN barely moves the guidance ensemble (shift 0.12 and 0.06, CV 0), so its transfer ratio has a near-zero denominator and is not reported. Yet it matches the policy's in-lung LHR: an unguided translation between the labelled classes targets lesions within the lung as well as the guided policy does. Lesion targeting and classifier validity are separate properties, and a protocol needs both.

The generator's own default schedule without a policy (the "no policy" row of Table E3) shows the same validity–transfer pattern as the policy: n→p CV 0.940 against a 3-validator mean transfer of 0.307 and `nih` 0.101.

E4. Lock-width sensitivity (exploratory)

The width of the background lock's editable band (main paper Sec. 3.2: dilation and Gaussian feathering of the lung mask, both in latent cells) was set once and not tuned. As an exploratory check we regenerated the first 50 images of each direction with the default policies under three settings: d0 (dilation 0, no feathering), d2 (dilation 2, feathering 1.0; the default setting, so a rerun) and d4 (dilation 4, feathering 2.0). LHR here is the whole-image lift, on the 47 annotated p→n images. Table E2 gives each setting's paired difference from the d2 rerun, with 95% paired percentile-bootstrap CIs (10,000 resamples of the images, drawn jointly across settings; transfer as a ratio of means; CV also with an exact McNemar test). As the rerun-noise reference we use the difference between the d2 rerun and the earlier evaluation of the default on the same images, and count a difference as exceeding rerun noise when its CI lies entirely outside ± that reference. The reference itself has CIs excluding zero on several metrics, so a CI that excludes zero is not on its own evidence of a setting effect. Every contrast here is across GPU types, and for p→n metrics that reference evaluation used the n→p adapter for both directions, so its band also contains that change.

Narrowing the lock (d0) raises edit concentration ($+0.028$ n→p, $+0.035$ p→n) and lowers the fraction of pixels changed (-0.060, -0.065); it also moves two n→p validator ratios beyond noise, in opposite directions (`nih` -0.059, `pc` $+0.019$). Widening it (d4) lowers whole-image LHR lift by 0.130 (95% CI $[0.038, 0.247]$), which a wider band that lets edit mass leave the lung would produce without any change in targeting (Supp. E7; we did not recompute this exploratory check in-lung), and makes edits slightly more diffuse (concentration -0.007 and -0.018, fraction changed $+0.018$ and $+0.048$). The other differences, including CV and mean transfer in both directions, are within rerun noise; d0's p→n CV drop of 0.140 (1 vs. 8 discordant images, $p = 0.039$) has a CI that excludes zero but not the noise band. With 50 images and a single rerun pair as the noise reference, we read these as indications of sensitivity, not as a tuning result.

E5. Same-image runs: default, no policy, fixed schedules, second seed

Table E3 compares five runs on the same 400 test images with the same per-image seeds: the default policy, the generator's default schedule without a policy, a fixed schedule that holds every control at the policy's mean action over its evaluation steps (same step budget and validity-based stopping, no training), a tuned fixed schedule chosen without RL, and a second training seed of the default configuration (LoRA and PPO base seed 1). Each direction uses its own adapter, except the second seed, whose two directions share one; both of its directions were resumed from checkpoints after preemption (n→p at update 15, p→n at update 80) without saved random-number state. The fixed schedule's raw actions are, for n→p, (-9.114, -0.671, -2.007, 2.446, 3.041, -6.357, 4.074, 6.550, -1.090) (strength 0.200, three passes, guidance bias 1.299), and for p→n, (1.705, 1.700, 0.432, 0.486, -1.607, -0.260, -1.142, 2.968, -0.608), the means over 201 and 1,093 policy steps. The tuned fixed schedule is selected on training-fold images only, by the quantity PPO maximizes, the mean discounted episode reward. Candidates are 54 constant schedules on a coarse grid (strength $\{0.25, 0.45, 0.65\}$, classifier scale $\{0.50, 0.85, 1.15\}$, guidance bias $\{0.85, 1.05, 1.27\}$, two or three passes; every other control at the midpoint of its range) and a local grid around the policy's mean action (three levels each of strength, classifier and preservation scale, strength boost and, for n→p, ramp start: 243 schedules for n→p, 81 for p→n). Each is scored on 12 training-

Table E2. Lock-width sensitivity, first 50 images per direction (LHR: 47 annotated). d2 is the default setting, rerun. Differences are setting minus d2 rerun, with 95% paired bootstrap CIs. Rerun noise is d2 rerun minus the earlier default evaluation on the same images. †CI entirely outside ± rerun noise. ‡CI excludes zero but not the rerun-noise band.

Metric	d2 (rerun)	Rerun noise	d0 − d2	d4 − d2
CV, n→p	0.960	0.000 [0.000, 0.000]	−0.020 [−0.060, 0.000]	0.000 [0.000, 0.000]
CV, p→n	0.740	−0.060 [−0.140, 0.000]	−0.140 [−0.260, −0.040]‡	+0.040 [−0.040, +0.120]
LHR, whole image (p→n)	2.489	−0.034 [−0.114, +0.043]	+0.013 [−0.096, +0.124]	−0.130 [−0.247, −0.038]†
Edit conc., n→p	0.248	0.000 [−0.002, +0.001]	+0.028 [+0.025, +0.032]†	−0.007 [−0.010, −0.004]†
Edit conc., p→n	0.317	−0.005 [−0.013, +0.002]	+0.035 [+0.025, +0.043]†	−0.018 [−0.025, −0.010]†
Frac. changed, n→p	0.664	0.000 [0.000, +0.001]	−0.060 [−0.072, −0.050]†	+0.018 [+0.008, +0.025]†
Frac. changed, p→n	0.629	+0.001 [−0.008, +0.009]	−0.065 [−0.074, −0.056]†	+0.048 [+0.037, +0.060]†
Transfer mean, n→p	0.262	−0.005 [−0.015, +0.002]	−0.016 [−0.032, −0.001]‡	+0.004 [−0.007, +0.015]
Transfer mean, p→n	0.251	−0.038 [−0.062, −0.014]	−0.025 [−0.052, +0.003]	+0.046 [+0.015, +0.077]‡
nih, n→p	0.042	−0.012 [−0.040, +0.010]	−0.059 [−0.101, −0.018]†	+0.017 [−0.010, +0.047]
pc, n→p	0.442	−0.002 [−0.007, +0.004]	+0.019 [+0.006, +0.031]†	−0.002 [−0.018, +0.012]
mimic_ch, n→p	0.304	−0.002 [−0.006, +0.002]	−0.008 [−0.024, +0.007]	−0.004 [−0.013, +0.006]
nih, p→n	0.124	−0.060 [−0.105, −0.021]	−0.038 [−0.082, +0.007]	+0.073 [+0.018, +0.129]‡
pc, p→n	0.320	−0.004 [−0.040, +0.030]	−0.029 [−0.070, +0.010]	+0.033 [+0.001, +0.068]‡
mimic_ch, p→n	0.307	−0.049 [−0.097, −0.006]	−0.009 [−0.061, +0.045]	+0.031 [−0.008, +0.070]

fold images per direction, and the best of each grid on 36 more; the stop bonus that the coarse grid's neutral stop threshold forgoes is added back before ranking. Both selected schedules come from the coarse grid: strength 0.25, classifier scale 1.15, guidance bias 0.85 and two passes for n→p, and strength 0.25, classifier scale 1.15, guidance bias 1.27 and three passes for p→n.

Against the default policy (paired, in-lung LHR on the 148 annotated p→n images with whole-image differences in parentheses, CV with exact McNemar tests): the fixed schedule differs by -0.003 in LHR ($[-0.040, +0.030]$, $p = 0.86$; whole image -0.071, $p = 0.12$), 0.000 in n→p CV (0 discordant) and -0.010 in p→n CV (14 vs. 16 discordant, $p = 0.86$); the second seed by $+0.025$ ($[-0.017, +0.065]$, $p = 0.23$; whole image -0.117, $p = 0.039$), -0.030 (0 vs. 6, $p = 0.031$) and -0.175 (7 vs. 42, $p = 3.6 \times 10^{-7}$); no policy by -0.077 ($[-0.120, -0.037]$, $p = 5 \times 10^{-4}$; whole image -0.532), -0.005 (3 vs. 4, $p = 1.0$) and -0.445 (1 vs. 90, $p = 7.4 \times 10^{-26}$). Relative to no policy, the fixed schedule recovers 96% ($[54, 159]\%$) of the policy's in-lung LHR gain and 98% ($[86, 111]\%$) of its p→n CV gain (paired bootstrap over images, 10,000 resamples); in n→p the policy has no gain to recover. The tuned fixed schedule differs from the default policy by $+0.012$ in LHR ($[-0.023, +0.047]$, $p = 0.51$; whole image -0.187, $p = 1.0 \times 10^{-4}$), -0.005 in n→p CV (1 vs. 2 discordant, $p = 1.0$) and $+0.050$ in p→n CV (22 vs. 12, $p = 0.12$); from the mean-action schedule by $+0.015$ in LHR ($[-0.023, +0.053]$, $p = 0.45$) and $+0.060$ in p→n CV (21 vs. 9, $p = 0.043$); and from no policy by $+0.088$ in LHR ($[+0.051, +0.127]$, $p = 1.1 \times 10^{-5}$). Its mean p→n transfer is below that of the policy and of the mean-action schedule (paired differences -0.065 $[-0.085, -0.045]$ and -0.032 $[-0.052, -0.013]$); in n→p the three schedules differ by at most 0.005. A search over fixed schedules by the training reward thus matches the policy's validity and in-lung lesion targeting but not its transfer, which the reward does not measure. Its whole-image LHR deficit reflects less edit mass in the lung (61% vs. 66%), not worse targeting within it.

E6. Edit locality and FID uncertainty

Table E4 is computed offline from the saved counterfactuals of the first three runs of Table E3 (no regeneration). With $\delta = |x_c - x|$ on the 256^2 grid: the share of $\sum \delta$ inside the editable region ($m > 0.5$, where m is the background lock's soft mask of main-paper Sec. 3.2, recomputed from the source image and upsampled bilinearly) and inside the undilated lung mask ($S > 0.5$), each beside its area fraction, which is what a spatially uniform edit would score; the fraction of pixels with $\delta > 0.01$, the same fraction measured against the VAE round trip $D(E(x))$ (posterior mean), and the fraction the round trip alone changes; and the share of $|x_c - D(E(x))|$ outside $m > 0.5$. FID uses the main paper's features, with a 95% interval from 500 bootstrap resamples of both sets; KID uses the same features (cubic polynomial kernel, unbiased, 100 subsets of 100). The recomputed FIDs match the evaluation's to within 0.01.

Three observations. (i) The lock works as intended at the level it acts on: relative to the VAE round trip, only 3–7% of the change lies outside the editable region. Most of the difference outside it, and about half of all pixels changed by more than 0.01, is VAE reconstruction error that any latent-space editor with this autoencoder incurs, so the main paper's "fraction of pixels changed" overstates the generative edit; against $D(E(x))$ it is 0.41–0.44 for the default policy. (ii) The edits are lung-localized but diffuse within the lung: about 82% of the mass sits in a region covering 49–59% of

Table E3. Five runs on the same 400 test images and seeds. CV with Wilson 95% CIs; LHR (in-lung, and whole-image for reference) with a t-interval (concentration and fraction changed have t-interval half-widths ≤ 0.01); transfer per validator with a 10,000-resample bootstrap, the mean with a paired bootstrap over per-image deltas. †Second seed: shared adapter, resumed training.

Dir.	Run	CV	LHR in lung	LHR whole image	nih	pc	mimic_ch	Transfer mean	Edit conc.	Frac. changed
n→p	default policy	0.945 [0.904, 0.969]	—	—	0.044 [0.005, 0.082]	0.465 [0.436, 0.494]	0.333 [0.302, 0.364]	0.281 [0.259, 0.303]	0.254	0.647
	no policy	0.940 [0.898, 0.965]	—	—	0.101 [0.059, 0.142]	0.474 [0.444, 0.503]	0.345 [0.314, 0.376]	0.307 [0.284, 0.329]	0.268	0.641
	fixed schedule	0.945 [0.904, 0.969]	—	—	0.044 [0.004, 0.083]	0.469 [0.439, 0.499]	0.334 [0.302, 0.365]	0.282 [0.260, 0.304]	0.254	0.647
	tuned fixed schedule	0.940 [0.898, 0.965]	—	—	0.039 [0.002, 0.077]	0.466 [0.437, 0.494]	0.326 [0.294, 0.358]	0.277 [0.256, 0.299]	0.252	0.637
	second seed†	0.915 [0.868, 0.946]	—	—	0.204 [0.161, 0.246]	0.458 [0.428, 0.488]	0.334 [0.304, 0.365]	0.332 [0.310, 0.355]	0.279	0.675
p→n	default policy	0.670 [0.602, 0.731]	1.165 [1.123, 1.207]	2.595 [2.450, 2.741]	0.165 [0.138, 0.193]	0.306 [0.278, 0.335]	0.357 [0.317, 0.397]	0.276 [0.254, 0.298]	0.314	0.634
	no policy	0.225 [0.173, 0.288]	1.088 [1.054, 1.122]	2.064 [1.961, 2.166]	0.113 [0.083, 0.143]	0.245 [0.218, 0.275]	0.149 [0.099, 0.197]	0.169 [0.146, 0.192]	0.278	0.599
	fixed schedule	0.660 [0.592, 0.722]	1.162 [1.118, 1.205]	2.525 [2.385, 2.664]	0.151 [0.124, 0.179]	0.261 [0.236, 0.288]	0.319 [0.278, 0.357]	0.244 [0.224, 0.264]	0.312	0.631
	tuned fixed schedule	0.720 [0.654, 0.778]	1.177 [1.143, 1.210]	2.408 [2.303, 2.514]	0.180 [0.151, 0.210]	0.205 [0.180, 0.230]	0.249 [0.208, 0.290]	0.211 [0.190, 0.232]	0.305	0.637
	second seed†	0.495 [0.426, 0.564]	1.190 [1.151, 1.229]	2.479 [2.350, 2.608]	0.154 [0.126, 0.182]	0.218 [0.192, 0.246]	0.252 [0.208, 0.295]	0.208 [0.187, 0.230]	0.298	0.620

Table E4. Edit locality and distribution metrics, $n = 200$ per direction and run. "in m" and "in lung": share of $\sum |x_c - x|$ inside the editable region and the lung mask, with the area fraction (uniform-edit null) in parentheses. "changed": fraction of pixels with $|\cdot| > 0.01$ against x, against $D(E(x))$, and for $D(E(x))$ itself. "out of m": share of $\sum |x_c - D(E(x))|$ outside the editable region. Locality values have 95% t-interval half-widths ≤ 0.017. KID $\times 10^3$, mean $\pm$ s.d. over subsets.

Dir.	Run	in m (null)	in lung (null)	changed vs. x	changed vs. $D(E(x))$	$D(E(x))$ floor	out of m vs. $D(E(x))$	FID [95% CI]	KID$\times 10^3$
n→p	default policy	0.835 (0.590)	0.653 (0.339)	0.647	0.436	0.499	0.036	22.39 [20.43, 24.56]	69.3 ± 3.6
	no policy	0.845 (0.590)	0.675 (0.339)	0.641	0.429	0.499	0.031	15.92 [14.33, 17.78]	48.4 ± 3.1
	fixed schedule	0.836 (0.590)	0.655 (0.339)	0.647	0.434	0.499	0.034	22.51 [20.78, 24.49]	70.0 ± 3.2
p→n	default policy	0.808 (0.495)	0.663 (0.268)	0.634	0.406	0.474	0.054	4.76 [3.86, 5.87]	12.5 ± 1.4
	no policy	0.734 (0.495)	0.558 (0.268)	0.599	0.359	0.474	0.068	2.69 [2.31, 3.24]	5.1 ± 0.5
	fixed schedule	0.798 (0.495)	0.647 (0.268)	0.631	0.402	0.474	0.058	4.40 [3.61, 5.27]	11.6 ± 1.4

the image, while edit concentration stays at 0.25–0.31. In p→n the policy places more of the change in the lung than the no-policy schedule (0.663 vs. 0.558, null 0.268), consistent with its LHR gain, and the fixed schedule is close to the policy (0.647). (iii) FID and KID favor the no-policy schedule in both directions, with non-overlapping bootstrap intervals, while the policy and the fixed schedule are indistinguishable. Since the feature extractor also guides generation (main-paper Sec. 4.1), we read this only as evidence that the stronger schedule moves counterfactuals further in the guidance classifier's feature space, not as a realism ranking.

E7. Lesion Hit Rate under alternative nulls

Main-paper Eq. (5) restricts both the edit mass and the area null to the lung mask L (lung probability > 0.5, the segmenter that drives the lock). Table E5 reports it next to the alternatives, recomputed offline from the saved counterfactuals of every run. The soft-lung variant weights by lung probability instead of thresholding it (1.240 for the default). The editable-region variant uses the lock's own soft mask m, which extends about 16 px beyond the lung. "vs. $D(E(x))$" measures the edit as $|x_c - D(E(x))|$, removing the VAE round trip. "Other boxes" scores each edit against every other annotated image's box (mean over the other 147) and estimates the lift explained by where lesions typically lie. Three lesion-blind references calibrate the whole-image null: a uniform wash over the image scores 1.00, over the editable region 1.90 [1.85, 1.95], and over the lung 3.21 [3.07, 3.34]. The VAE round trip alone, $|D(E(x)) - x|$, scores 1.06 on the whole image and 0.98 [0.97, 1.00] in the lung, so autoencoder error carries no lesion signal.

Paired in-lung differences between runs (148 images unless noted; bootstrap 95% CI, paired t-test): default minus no policy $+0.077$ $[+0.037, +0.120]$, $p = 5 \times 10^{-4}$ (whole image $+0.532$); minus fixed schedule $+0.003$, $p = 0.86$; minus tuned fixed schedule -0.012, $p = 0.51$ (whole image $+0.187$, $p = 1.0 \times 10^{-4}$); minus second seed -0.025, $p = 0.23$; minus single-classifier on the same images $+0.082$ $[+0.042, +0.124]$, $p = 2 \times 10^{-4}$; minus Grad-CAM $+0.025$ $[-0.031, +0.080]$, $p = 0.37$ (whole image $+0.960$); minus Latent Shift (iterative) $+0.059$ $[+0.020, +0.099]$, $p = 0.004$; minus CycleGAN $+0.005$, $p = 0.84$. Under matched adapters (default evaluated with the n→p adapter for both directions): minus sparsity $+0.026$ $[-0.003, +0.056]$, $p = 0.085$; minus single-classifier on the 76 shared images $+0.074$ $[+0.022, +0.125]$, $p = 0.006$. Single-classifier minus random actions $+0.003$ $[-0.024, +0.031]$, $p = 0.85$ (whole image $+0.090$, $p = 0.012$). The own-box minus other-boxes difference for the default is $+0.076$ $[+0.037, +0.117]$, $p = 3 \times 10^{-4}$, the lesion-specific part of its in-lung lift.

E8. Calibrating the transfer ratio

Validators on real images. Table E6 scores every validator output on the 5,337-image test fold against RSNA's labels. The real-pair ceiling is ρ_v computed between popu-

Table E5. LHR (p→n) under alternative nulls, mean over annotated images (in-lung with 95% t-interval). In-lung share: fraction of $\sum|x_c - x|$ inside L. *Own classifier-selected groups (single-classifier and random: 183 annotated; 3-model: 27). All other rows use the default's 148 annotated images.

Run	Whole image	**In lung**	Editable region m	In lung, vs. $D(E(x))$	In lung, other boxes	In-lung share
Default policy	2.595	**1.165** [1.123, 1.207]	1.660	1.163	1.089	0.661
No policy	2.064	1.088 [1.054, 1.122]	1.447	1.082	1.043	0.556
Fixed schedule (policy mean)	2.525	1.162 [1.118, 1.205]	1.634	1.161	1.071	0.643
Tuned fixed schedule	2.408	1.177 [1.143, 1.210]	1.612	1.179	1.080	0.613
Second seed	2.479	1.190 [1.151, 1.229]	1.644	1.191	1.094	0.617
Single-classifier, default's images	2.296	1.083 [1.044, 1.122]	1.502	1.087	1.023	0.625
Single-classifier*	2.293	1.084 [1.048, 1.120]	1.507	1.086	1.020	0.643
Random actions*	2.204	1.081 [1.044, 1.119]	1.483	1.085	1.021	0.618
3-model ensemble*	2.492	1.216 [1.125, 1.307]	1.626	1.233	1.017	0.633
Sparsity reward	2.398	1.167 [1.131, 1.204]	1.605	1.170	1.075	0.610
Grad-CAM constraint	1.636	1.140 [1.089, 1.191]	1.368	1.280	1.078	0.408
Latent Shift (iterative)	1.451	1.106 [1.077, 1.135]	1.257	1.103	1.043	0.367
CycleGAN	1.483	1.160 [1.117, 1.203]	1.339	1.068	1.096	0.359
Uniform wash over L	3.206	1	—	—	—	1
VAE round trip only	1.058	0.982 [0.968, 0.995]	—	—	—	—

Table E6. Validator outputs on the test fold: AUROC, real-pair ceiling ρ [95% CI], and ρ_v of the default policy's counterfactuals (n→p / p→n).

Validator	Output	AUROC	Ceiling	Default ρ_v
`nih`	Pneumonia	0.766	0.47 [0.44, 0.49]	0.04 / 0.17
	Consolidation	0.802	0.34 [0.33, 0.36]	0.16 / 0.09
	Infiltration	0.740	0.26 [0.24, 0.28]	0.10 / 0.09
`pc`	Pneumonia	0.783	0.58 [0.56, 0.61]	0.47 / 0.31
	Consolidation	0.798	0.61 [0.59, 0.63]	0.49 / 0.25
	Infiltration	0.804	0.77 [0.74, 0.79]	0.51 / 0.38
`mimic_ch`	Pneumonia	0.728	0.42 [0.40, 0.45]	0.33 / 0.36
	Consolidation	0.720	0.45 [0.43, 0.48]	0.37 / 0.35
	Lung Opacity	0.770	0.63 [0.60, 0.65]	0.38 / 0.41

lations: the validator's mean score on positives minus negatives, over the same difference for the guidance ensemble (AUROC 0.872), with a bootstrap over images resampled within each class. The Youden threshold maximizes sensitivity plus specificity on the same fold; it defines each validator's decision for the flip rates below. RSNA's positive class is "Lung Opacity" and NIH's Pneumonia label is rare, so we also score the Consolidation, Infiltration and Lung Opacity outputs where a validator has them.

The semantically closest outputs do not change the picture. `mimic_ch`'s Lung Opacity output is its best (AUROC 0.770) and its ρ_v is higher by 0.05, in proportion to its higher ceiling. `nih`'s Consolidation output transfers more than its Pneumonia output in n→p (0.16 vs. 0.04), but still well below its ceiling.

Attack floor. On the default's 200 + 200 source images we attack the guidance ensemble alone, in pixel space $[0, 1]$ at 256^2, with the loss $-\log f$ toward positive or $-\log(1 - f)$ toward negative: FGSM at $\epsilon \in \{1, 2, 4, 8\}/255$, PGD with 10 steps of $\epsilon/4$ at $\epsilon \in \{2, 4, 8\}/255$ (no random start), and L_2 PGD with 10 steps whose per-image budget is that image's $\|x_c - x\|_2$ (mean 12.4 n→p, 17.6 p→n; clipping to $[0, 1]$ leaves 9.1 and 13.4). The adversarial images are scored by exactly the code that scores the counterfactuals.

The two directions differ. Removing evidence of opacity transfers poorly when done adversarially: every PGD L_∞ attack stays at or below 0.04 on `nih` and `pc` and 0.18 on `mimic_ch`, while the counterfactuals reach 0.17–0.36. (FGSM barely moves the guidance ensemble in this direction, so its ratios have small denominators.) Adding it transfers readily: attacks with a quarter of the counterfactual's norm already reach 0.22–0.31, and the norm-matched attack matches or exceeds the counterfactual on `pc` and `mimic_ch`. These DenseNet-121s share architecture and much of their training distribution, so shared adversarial directions are expected. The norm-matched L_2 attack concentrates its budget into large local changes (L_∞ 0.37 n→p, 0.55 p→n) and may itself synthesize opacity-like structure. It is therefore an upper bound on what an attack can reach at that norm, not a pure-perturbation floor.

Decision flips complement the ratio. At their own operating points, `pc` and `mimic_ch` flip 91% and 69% of n→p and 91% and 80% of p→n counterfactuals, far more than a ratio near 0.3 suggests: the ratio averages score shifts, while a flip only needs enough shift to cross the threshold. Prevalence-matched thresholds (the top 22.4% of test images called positive) sit higher and give the same ordering. In p→n the counterfactuals flip 79%, 100% and 89% against 17%, 29% and 48% for PGD L_∞ 4/255. In n→p they flip 6%, 54% and 33%, against 9%, 12% and 14% for that attack and 9%, 53% and 89% for the norm-matched one (`nih`, `pc`, `mimic_ch`).

Table E7. Transfer ratio and decision flips of attacks on the guidance ensemble, against the default policy's counterfactuals on the same source images. Guidance CV at $\tau = 0.8$ is the fraction reaching $f \geq 0.8$ (n→p) or $f \leq 0.2$ (p→n). Flips: share of sources on the source side of a validator's Youden threshold that end on the target side. Pneumonia outputs.

					ρ_v			Youden flips		
Dir.	Perturbation	Guidance Δ	$CV_{0.8}$	$\|\cdot\|_2$	nih	pc	mimic_ch	nih	pc	mimic_ch
n→p	**default counterfactual**	0.898	1.00	12.4	0.04	0.47	0.33	0.22	0.91	0.69
	FGSM 4/255	0.646	0.39	4.0	0.31	0.42	0.30	0.42	0.34	0.32
	FGSM 8/255	0.752	0.69	7.9	0.40	0.48	0.45	0.80	0.69	0.86
	PGD L_∞ 2/255	0.769	0.78	1.8	0.13	0.16	0.11	0.16	0.10	0.16
	PGD L_∞ 4/255	0.908	1.00	3.4	0.22	0.31	0.23	0.46	0.34	0.37
	PGD L_∞ 8/255	0.914	1.00	6.1	0.27	0.42	0.37	0.58	0.79	0.92
	PGD L_2, norm-matched	0.914	1.00	9.1	0.18	0.45	0.49	0.36	0.97	0.98
p→n	**default counterfactual**	0.761	0.84	17.6	0.17	0.31	0.36	0.48	0.91	0.80
	FGSM 8/255	0.231	0.00	7.9	0.01	0.24	0.40	0.07	0.27	0.31
	PGD L_∞ 2/255	0.643	0.47	1.7	0.01	0.02	0.03	0.04	0.03	0.04
	PGD L_∞ 4/255	0.821	0.96	3.2	0.02	0.02	0.07	0.07	0.07	0.16
	PGD L_∞ 8/255	0.837	0.99	5.7	0.04	0.04	0.18	0.17	0.16	0.47
	PGD L_2, norm-matched	0.836	0.99	13.4	0.07	0.46	0.02	0.35	0.99	0.23

E9. Exploratory: lesion targeting on mammograms

An earlier port of PPO-CF to full-field mammograms of CBIS-DDSM (Lee *et al*., Sci. Data 2017) offers what RSNA cannot: an anchor for the n→p direction. CBIS-DDSM contains no normal mammograms, so the classes are benign and malignant, and every source image in either direction carries radiologist ROI masks. The run is not part of the main results, for three reasons established by an audit of its saved counterfactuals. (i) Its guidance classifier, a DenseNet-121 we trained on CBIS-DDSM, had 336 of the 400 evaluation images (304 of 362 patients) in its training set, so its CV of 1.00 in both directions says little. (ii) No held-out validator we could build reaches AUROC 0.65 on real images: out-of-fold AUROC is 0.62–0.64 for ResNet-18 and EfficientNet-B0 validators trained on the 234 patients disjoint from the guidance model's training set, and for validators trained on CMMD, a separate mammography corpus. The transfer ratio is therefore uninterpretable and not reported. (iii) The generator and policy were trained before the split and seeding controls of main-paper Sec. 4.4, and no no-policy run exists on these images.

We therefore report only LHR, which needs neither a validator nor a clean classifier, recomputed offline from the saved counterfactuals (200 per direction). The image indices were recovered by replaying the group scan, and every source image was re-checked to be pixel-identical to the dataset image at that index. The lesion mask is the union of the image's ROI masks, resized exactly like the image. The organ mask B is the breast channel of the analytic segmenter (Otsu threshold and largest connected component) that gated the background lock in this run. LHR is main-paper Eq. (5) with B in place of the lung mask. Of 200 sources, 191 (benign→malignant) and 196 (malignant→benign) are kept: 12 ROI masks do not match their mammogram's size and one image has none.

Table E8. LHR on CBIS-DDSM counterfactuals (exploratory; see text for why). Mean over images with 95% t-interval; paired differences with a 10,000-resample bootstrap CI.

	benign→malignant ($n = 191$)	malignant→benign ($n = 196$)
In breast (Eq. 5 with B)	**1.547** [1.420, 1.674]	**1.892** [1.747, 2.038]
In breast, vs. $D(E(x))$	1.539	1.857
In breast, other images' ROIs	1.134	1.133
VAE round trip only, in breast	1.249 [1.196, 1.302]	1.276 [1.236, 1.315]
Whole image	4.053	4.373
Uniform wash over B, whole image	3.010	2.734
Edit mass inside B	0.88	0.86
Own minus other ROIs	+0.413 [+0.294, +0.547]	+0.760 [+0.615, +0.913]
Edit minus VAE round trip	+0.298 [+0.177, +0.425]	+0.617 [+0.483, +0.756]

Both directions put more edit mass on the lesion than a lesion-blind edit inside the breast, more strongly than on RSNA (1.55 and 1.89 against 1.16), and here the edits beat an in-organ wash even under the whole-image null. Unlike on RSNA, the VAE round trip is itself lesion-directed (1.25 and 1.28), plausibly because masses and calcifications are the high-contrast detail the autoencoder reconstructs worst. Measuring the edit against $D(E(x))$ nevertheless leaves LHR almost unchanged, and the edits exceed the round trip by 0.30 and 0.62. Most of the lift is image-specific: other images' ROIs score 1.13. The benign→malignant direction, which has no RSNA analogue, targets the lesion as well. We do not read this as evidence that the policy or the lock produces the targeting, since no no-policy run exists. Nor is it evidence of class-relevant content, since the validators are uninformative and the guidance classifier is contaminated. It shows that the in-organ LHR and its nulls transfer to a second modality, where they also give an answer in both directions.

E10. PPO seed replicates

To give the ensemble-versus-single-classifier comparison a run-to-run check, we retrained the PPO policies of both configurations with new seeds while keeping every other component fixed: each direction loads the LoRA adapter trained for it in that configuration's original run (checked by hash), so only the PPO seed changes. With the base seed b, the n→p and p→n policies use seeds $b+1$ and $b+2$; the original runs have $b=0$, the replicates $b=1$ and $b=2$ for the single classifier and $b=2$ for the ensemble. Each policy is trained for 800 episodes per direction. The three p→n replicate runs were preempted once and resumed from their last checkpoint (at updates 70, 75 and 20 of 100) without saved random-number state; the n→p runs were not interrupted. With the earlier second training seed of the default (Supp. E5; $b=1$ for LoRA and PPO, one retrained adapter shared by both directions, both directions resumed), this gives three ensemble and three single-classifier runs. All are evaluated on the default's 400 test images with the same per-image seeds, so every comparison below is paired. CV is computed with each run's own guidance classifier (the ensemble, or `rsna` alone), on the default's classifier-selected groups.

Table E9. PPO seed replicates on the default's 400 test images. CV with Wilson 95% CI; transfer is the mean ρ_v over the three validators; in-lung LHR with t-interval on the 148 annotated images. †LoRA also retrained (seed 1), one adapter for both directions; resumed training (Supp. E5).

Guidance	b	CV n→p / p→n	Transfer n→p / p→n	LHR in lung	Whole
Ensemble	0	0.945 / 0.670	0.281 / 0.276	1.165 [1.123, 1.207]	2.595
	1†	0.915 / 0.495	0.332 / 0.208	1.190 [1.151, 1.229]	2.479
	2	0.950 / 0.765	0.269 / 0.229	1.140 [1.106, 1.175]	2.400
Single (`rsna`)	0	0.855 / 0.840	0.245 / 0.141	1.083 [1.044, 1.123]	2.296
	1	0.855 / 0.800	0.299 / 0.116	1.080 [1.046, 1.114]	2.220
	2	0.855 / 0.850	0.232 / 0.141	1.107 [1.069, 1.144]	2.356
No policy	—	0.940 / 0.225	0.307 / 0.169	1.088 [1.054, 1.122]	2.064

Wilson intervals for CV at $n=200$ have half-widths of 0.03 to 0.07 (*e.g.* $[0.910, 0.973]$ for 0.950 and $[0.702, 0.818]$ for 0.765). The three single-classifier n→p CVs coincide at 171 of 200, but the valid images and the counterfactuals differ between seeds.

Paired in-lung LHR differences (bootstrap 95% CI, paired t-test). Within a configuration, seeds do not differ detectably: ensemble $b=0$ minus $b=2$ $+0.025$ $[-0.007, +0.057]$, $p=0.14$, and $b=0$ minus $b=1$ -0.025, $p=0.23$. Across configurations, each of the nine ensemble-minus-single pairs is positive, from $+0.034$ ($b=2$ vs. $b=2$; $[-0.003, +0.072]$, $p=0.076$) to $+0.110$; the other eight have $p<0.01$. Averaging each image's lift over the three runs of each configuration gives $+0.075$ $[+0.050, +0.102]$ ($+0.063$ $[+0.037, +0.092]$ without the shared-adapter run). Against no policy, the ensemble runs gain $+0.077$, $+0.102$ and $+0.052$ ($[+0.014, +0.091]$, $p=0.008$), and the single-classifier runs -0.005, -0.008 and $+0.019$ (all $p \geq 0.31$). Scored against other patients' boxes (Supp. E7), the lesion-specific part of the lift is $+0.072$ $[+0.038, +0.105]$ for the $b=2$ ensemble run and $+0.057$ to $+0.081$ for the single-classifier runs.

What survives seed variation: the ensemble's advantage in in-lung LHR and in p→n transfer, which hold for every pair of runs; the ensemble policy's gain over no policy in in-lung LHR and p→n CV; and the single-classifier policy's lack of one. In n→p every run combines high CV with transfer near the default's (CV 0.855–0.950, transfer 0.23–0.33). What does not survive: an ensemble advantage in n→p transfer (the ranges overlap, and the $b=1$ single-classifier run exceeds the default). p→n CV depends on the seed as much as on the configuration; within the ensemble it spans 0.495–0.765, wider than the gap between the two configurations' means.